

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions x 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Bestha Akshaya	Reg. No.:	192424336
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Multimedia Rendering Engine

A multimedia rendering engine evaluates three algorithms with complexities $O(n)$, $O(n \log n)$, and $O(n^2)$. Analyze each algorithm using asymptotic notation and compare their growth rates. Explain how performance differs when processing high-resolution media files. Identify the most efficient algorithm for large-scale rendering tasks. Justify your answer based on computational efficiency and scalability. ###

Q2

Application: Social Media Feed Ranking

A social media platform ranks posts using: $T(n) = 4T(n/2) + n^2$. Apply the Master Theorem to compute the time complexity. Clearly specify parameters and case selection. Analyze scalability issues as the number of posts increases and interpret efficiency.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1) Application: Multimedia Rendering Engine

A multimedia rendering engine processes images and videos. The efficiency of the algorithm directly affects the rendering speed, especially when working with high resolution media files. Three algorithms are considered with time complexities $O(n)$, $O(n \log n)$, and $O(n^2)$.

Analysis

1. $O(n)$ Algorithm

- The running time increases linearly with the input size.

- If the input doubles, the execution time also approximately doubles.

- It is highly efficient & suitable for processing large multimedia files.

2. $O(n \log n)$ Algorithm

- The execution time grows slightly faster than linear.

- Many efficient algorithms like merge sort and heap sort have this complexity.

3. $O(n^2)$ Algorithm

- The running time increases quadratically
- It is suitable only for small datasets & becomes inefficient for high-resolution.

Comparison table

Algorithm	Time complexity	performance
Algorithm 1	$O(n)$	Best
Algorithm 2	$O(n \log n)$	Good
Algorithm 3	$O(n^2)$	Poor

Conclusion:

Among the three algorithms, $O(n)$ is the most efficient because it requires the least execution time as the input size increases. $O(n \log n)$ is also efficient and widely used for large scale. $O(n^2)$ is not suitable for rendering.

2) Application: social Media feed Ranking

Given:

$$T(n) = 4T(n/2) + n^2$$

we apply the master theorem to determine the time complexity.

Formula

$$T(n) = aT(n/b) + f(n)$$

Here

n = size of the problem

a, b = size of the sub problem

$f(n)$ = cost of a problem

Step-1:

Identify the

- $a = 4$

- $b = 2$

- $f(n) = n^2$

Step-2:

Calculate

$$n \log_b a = n \log_2 4 = n^2$$

$$f(n) = n^2 = n \log_b a$$

since both are equal,

$$T(n) = O(n \log_b^a) \text{ or}$$

$$p < -10(n^k)$$

this satisfies case 2 of master theorem.

Step-3: Apply Master Theorem

$$T(n) = O(n^2 \log n)$$

Scalability Analysis:

- As the no of social media posts increases, the running time also increases.
- the algorithm performs better than cubic-time algorithm.

Conclusion:

using master theorem,

$$T(n) = O(n^2 \log n)$$